

Amirhossein Donyadidegan

Junior Geoinformatics Engineer | GIS | Python | Remote Sensing | Spatial Data

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Profile

MSc Geoinformatics Engineering graduate with hands-on experience in GIS, remote sensing, Python development, geospatial analysis, interactive dashboards, and machine-learning workflows for spatial data.

Experience

GIS & Data Analyst, Research Assistant | IIP, Karlsruhe Institute of Technology (KIT) | Jul 2025 - Mar 2026

- Developed Python-based data pipelines for energy and mobility data processing.
- Performed spatial analyses supporting energy-demand and decarbonization models.
- Built interactive dashboards with map integration for applied research workflows.

Web Developer, Research Assistant | KIT nova | Jul 2025 - Dec 2025

- Contributed to web applications with focus on functionality and usability.
- Supported VR/AR projects for digital and interactive applications.
- Contributed to geospatial visualizations and digital map solutions.

Intern | Naghsheh Gostaran Fartak Co. | May 2021 - Sep 2021

- Supported surveying, photogrammetry, and GIS projects in a practical work environment.
- Assisted with data collection, spatial analysis, and technical implementation.
- Applied AutoCAD and GIS tools for creating and processing geospatial data.

Education

MSc Geoinformatics Engineering | Politecnico di Milano, Italy | Sep 2023 - Mar 2026 | Completed | Grade: 102 / 110

Exchange student in geodesy at the University of Bonn and in remote sensing and geoinformation at KIT. Thesis: Inferring Map Generalization Operations from User Prompts.

BSc Surveying Engineering | University of Tehran, Iran | Sep 2018 - Jul 2022 | Grade: 16.5 / 20

Thesis: Application of GIS and Big Data in Smart Cities.

Selected Projects

Inferring Map Generalization Operations from User Prompts - MSc thesis project using Python, Scikit-learn, GeoPandas, and Shapely.

LayerIterator - Python tool for rule-based raster modification using vector masks, Rasterio, GeoPandas, and NumPy.

LandsatToolkit - Reusable Python toolkit for Landsat satellite image processing and environmental analysis workflows.

AI-Based Landslide Susceptibility Mapping - GIS and machine-learning workflow using remote sensing, terrain, infrastructure data, and WebGIS.

SE4G Geospatial Data Visualization Dashboard - Interactive geospatial dashboard using Python, Dash, Plotly, and Flask.

Skills

Programming: Python, SQL, JavaScript, C++

GIS & Geospatial: QGIS, ArcGIS, GeoPandas, PostGIS, Google Earth Engine

Data & ML: Scikit-learn, PyTorch, Pandas, NumPy, Machine Learning

Visualization: Dash, Plotly, Matplotlib

Web & APIs: Flask, REST APIs, HTML, CSS

Languages: English fluent, German intermediate, Italian intermediate